Tarun Sai Mandava

Data Scientist

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SUMMARY

- Data Scientist with 4+ years of experience in data science and analytics, adept at leveraging machine learning, data mining, and statistical analysis to drive business solutions.
- Expertise in the full software development life cycle (SDLC) and Agile and Scrum methodologies.
- Proficient in utilizing various machine learning algorithms, including logistic regression, random forest, XGBoost, KNN, SVM, neural networks, linear regression, lasso regression, and k-means clustering.
- Skilled in advanced regression modeling, correlation, multivariate analysis, model building, and the application of statistical concepts.
- Experienced in predictive modeling, data mining methods, factor analysis, ANOVA, hypothesis testing, and other advanced statistical and econometric techniques.
- Proficient in Python, leveraging libraries such as Scikit-Learn, Matplotlib, Numpy, Scipy, and Pandas for data manipulation and analysis.
- Expert in data visualization tools, including Tableau and R-Shiny, creating impactful, interactive reports and dashboards to extract actionable insights from large datasets.
- Experience with relational and non-relational databases such as MySQL, SQLite, and SQL.
- Proven ability to parse, manipulate, and prepare data through advanced techniques such as regex, reindexing, melting, and reshaping.

SKILLS

Methodologies: SDLC, Agile, Waterfall
Language: Python, R, SQL, SAS, C#, Scala
IDEs: Visual Studio Code, PyCharm, Jupiter Notebook
Statistical Methods: Hypothetical Testing, ANOVA, Time Series
Machine Learning: Regression analysis, Bayesian Method, Decision Tree, Random Forests, Support Vector Machine, Neural Network, Sentiment Analysis, K-Means, KNN, Classification, SVM, Naive Bayes, NLP, LLM, CNN, XGBoost, Julia, Deep Learning, Predictive Models
Packages: NumPy, Pandas, Matplotlib, SciPy, ggplot2, Scikit-Learn, PyTorch, TensorFlow, Keras, Spark
Visualization Tools: Tableau, Power BI, Microsoft Excel
Cloud Technologies: AWS, GCP
Database: MySQL, SQL Server, Oracle, MongoDB
Software/Other Skills: Jira, Data Cleaning, Data Wrangling, Critical Thinking, Communication Skills, Presentation Skills, Problem-solving, Decision-Making, EDA, Communication Skills, Databricks, Data Visualization, Predictive Analytics, Pattern Recognition, JMP, Data Integrity, Quantitative Data, Data Science, Statistics, Statistical Analysis, Data Analytics, Data Modeling, Big Query, Snowflake, Data Extraction, Data Loading, Data Mining, Data Reporting, Data Transformation
Operating System: Windows, Linux

EDUCATION

Master of Science in Computer Science
Kennesaw State University, Kennesaw, GA

EXPERIENCE

Intuit, USA | Data Scientist **Nov 2023 - Current**

- Worked in Agile environments to deliver iterative solutions and adapt to changing project requirements.
- Trained and optimized multiple machine learning models, including Logistic Regression, Random Forest, and SVM, improving Customer churn prediction accuracy by 15%.
- Utilized Python's data visualization libraries (Matplotlib, Seaborn) to create actionable insights, facilitating cross-team communication and decision-making.
- Worked on Statistical methods like data-driven Hypothesis Testing and A/B Testing to draw inferences, determine significance level and derived P-value, and evaluate the impact of various risk factors.
- Implemented and tested models on AWS EC2, optimizing algorithms and parameters in collaboration with the development team, resulting in a 25% increase in model performance.
- Designed and developed interactive dashboards with Tableau and generated comprehensive reports, enhancing the team’s ability to interpret and act on data insights.
- Managed CI/CD pipelines for seamless deployment of machine learning models, reducing deployment time by 40%.
- Integrated machine learning models with RESTful APIs, enabling real-time predictions and improving system responsiveness.
- Utilized NLTK library for NLP data processing, identifying key patterns and insights that contributed to a 20% improvement in model accuracy.
- Crafted customized SQL queries using MS SQL Management Studio and Crystal Reports, streamlining data retrieval and report generation processes.

Maruti Techlabs, India | Data Scientist **Jan 2021 - Nov 2022**

- Engineered and cleansed large datasets using Pandas and NumPy, ensuring data consistency, integrity, and quality for downstream analysis.
- Developed and implemented predictive models (Naïve Bayes, Logistic Regression, Random Forest, SVM, XGBoost) to accurately predict loan defaults.
- Constructed ensemble models (Ridge, Lasso Regression, XGBoost) to estimate potential loan default losses.
- Utilized Spark (Pyspark, Spark SQL, MLlib) on AWS to conduct real-time analysis of loan default patterns.
- Created optimized SQL queries in Teradata SQL Workbench to support Tableau dashboards, providing actionable insights.
- Employed supervised and unsupervised learning techniques, including LLM (Latent Dirichlet Allocation), to uncover hidden patterns and insights within complex datasets.

Intex Technologies, India | Data Scientist **May 2019 - Dec 2020**

- Conducted thorough testing and validation of data science models and solutions before deployment, following a structured and rigorous approach typical of Waterfall methodologies.
- Engaged in Business and Data Analysis, Data Profiling, Data Migration, Data Integration, and Metadata Management Services, ensuring comprehensive data management and accuracy.
- Applied advanced machine learning algorithms, including PCA, K-nearest neighbors, random forest, gradient boosting, neural networks, and XGBoost, to accurately predict Higgs Boson weights and labels.
- Collaborated with cross-functional teams to integrate Databricks solutions into existing data pipelines, ensuring a seamless data flow.
- Developed and ingested data into an AWS cluster (EC2 instances) using SparkSQL, enhancing data handling capabilities.
- Developed deep learning models for industrial decision-making through Keras, implementing models such as RNN and LSTM.
- Implemented backup and recovery strategies for SQL Server databases, ensuring data availability and integrity in case of system failures or disasters.
- Utilized Natural Language Processing (NLP) for response modeling and fraud detection efforts for credit cards.
- Used pandas, NumPy, seaborn, SciPy, Matplotlib, scikit-learn, and NLTK in Python to develop various machine learning algorithms.
- Performed data manipulation and aggregation from different sources using Nexus, Toad, Business Objects, Power BI, and Smart View.
- Implemented classification algorithms using supervised techniques such as Logistic Regression, Decision Trees, KNN, and Naive Bayes.